Chapter 1: Introduction to Sabermetrics

Big hit. Big catch. Big pitch. If you've ever seen a baseball game, you understand that these simple events contribute immensely to the outcome of the game. But, what contributes most greatly to the outcome of the season? In my project, I plan to explore the effect that many common statistics have on predicting wins and more importantly, predicting championships. Then, I intend to try to discover what statistics determine a championship team.

Baseball statistics has gained much attention recently. Several books, such as *Moneyball*, explore the importance of drafting players based on certain statistics, rather than old-fashioned conventions [4]. In the game of baseball, in which no salary cap has been implemented, this kind of sabermetric approach is especially important to small market teams. Sabermetrics is defined as the analysis of baseball through objective evidence, especially statistics. The term is derived from the acronym SABR, which stands for the Society for American Baseball Research [10]. It was coined by Bill James, who was among its first proponents and has long been its most prominent and public advocate [1].

While many areas of study are still in development, it has yielded a number of interesting insights into the game of baseball and in the area of performance measurement. The following chapters will investigate some of these insights. In Chapter 2, I will explore linear and nonlinear methods to predict wins. Chapter 3 uses a simulation in order to gain perspective on the playoffs. Finally, Chapter 4 summarizes my research and leaves room for future work.

Chapter 2: Winning Percentage Prediction

2.1 Linear Winning Percentage Models

One of the simplest approaches when modeling data is linear regression. It has a variety of real world applications, thus, it seemed like a logical first attempt to model the Major League Baseball data. The model and its assumptions are discussed in the following sections [6].

2.1.1 Introduction to Linear Regression

The model for multiple linear regression is $Y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_p x_{ip} + \varepsilon_i$, where Y_i is the response variable and $x_{i1},...,x_{ip}$ are the predictors. $\beta_0,...,\beta_p$ are coefficients, and ε_i is a normally distributed error. We use the Least Squares Estimate (LSE) to determine $\hat{\beta}_0,...,\hat{\beta}_p$,which are the estimates of $\beta_0 .. \beta_p$. The LSE is found by minimizing the sum of squared residual error, or $\min s(\beta_0,...,\beta_p) = \sum_{i=1}^{n}(Y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip})^2$. Taking the partial derivatives with respect to each β_i, leads to the following normal equations:

$$
\begin{aligned}
\tfrac{\partial s}{\partial \beta_0} &= -2\sum_{i=1}^{n}(Y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip}) \\
\tfrac{\partial s}{\partial \beta_1} &= -2\sum_{i=1}^{n}x_{i1}(Y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip}) \\
\vdots \quad & \quad \vdots \qquad \vdots \\
\tfrac{\partial s}{\partial \beta_p} &= -2\sum_{i=1}^{n}x_{ip}(Y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip})
\end{aligned}
$$

Setting $\frac{\partial s}{\partial \beta_0} = \frac{\partial s}{\partial \beta_1} = ... = \frac{\partial s}{\partial \beta_p} = 0$ and solving, we get that the LSE of β as $\hat{\beta} = (X'X)^{-1}X'Y$ with covariance matrix $Cov(\hat{\beta}) = \sigma^2 (X'X)^{-1}$ [6].

In linear regression, there are some assumptions that are required before using the model. First of all, the predictors are assumed to be linearly independent. Therefore, they shouldn't be correlated with each other. Second, the residuals are assumed independent and identically distributed normal with mean 0 and variance σ^2 [6].

In order to check these assumptions, different plots may be used. Residuals plotted against the response variable, residuals plotted as a function of time, residuals plotted against the fitted values, and a normal probability plot of the residuals to test normality are all things which can be looked at to determine if linear regression is appropriate. The first 3 plots should look like a complete random diagram and the normal plot should look like a straight line [6].

2.1.2 Full Regression and Results

As an initial guess of what variables might be important, I chose 18 raw Major League Baseball statistics to predict wins (See Table 2.1). They represent the 3 categories of statistics: offensive, pitching, and defensive. See Appendix 5.1 for definitions of statistics.

Offensive	Pitching	Defensive
-Batting Average (AVG) -Hits (H) -On Base Percentage (OBP) -Runs Scored (R) -Slugging Percentage (SLG) -Stolen Bases (SB) -Strikeouts (SO) -Total Bases (TB) -Walks (BB)	-Earned Run Average (ERA) -Opponent Batting Averages (OAVG) -Runs Allowed (OR) -Saves (SV) -Strikeouts/Walk Ratio (K_BB) -Walks+Hits per Inning Pitched (WHIP)	-Defensive Efficiency Ratio (DER) -Errors (E) -Fielding Percentage (FPCT)

Table 2.1 Chosen Baseball Statistics

Using 1997-2006 data, a linear regression model was determined using SAS (Appendix 5.3). A brief analysis of the results revealed that the model fits the data relatively well with R^2 =0.9402 and Mean Square Error (*MSE*) of 9.27505. The normal plot, Figure 2.2, looks very promising as well. There seems to be a straight line that indicates that the data is normally distributed and can be well represented by a linear regression.

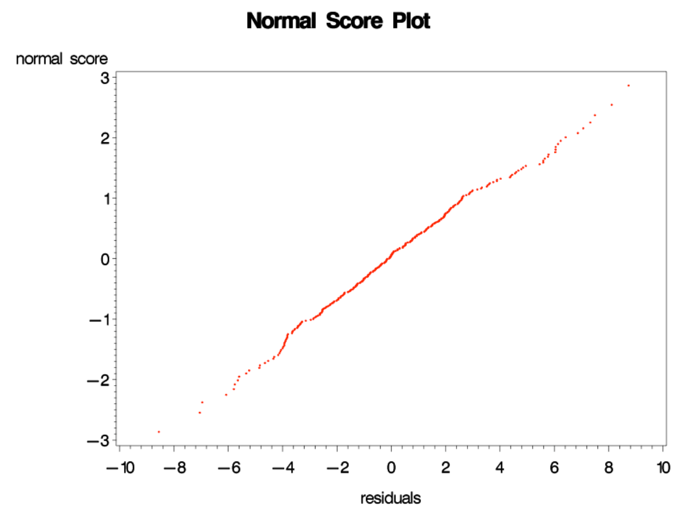

Figure 2.2 Normal score plot for 1997-2006 full regression

Overall, the residual plots look reasonable. With a few exceptions, the residuals seemed to be randomly distributed, which is another positive indication that linear regression is a good fit for this data. The time-order residual plot is shown in Figure 2.3, and the residual plot for runs scored is shown in Figure 2.4.

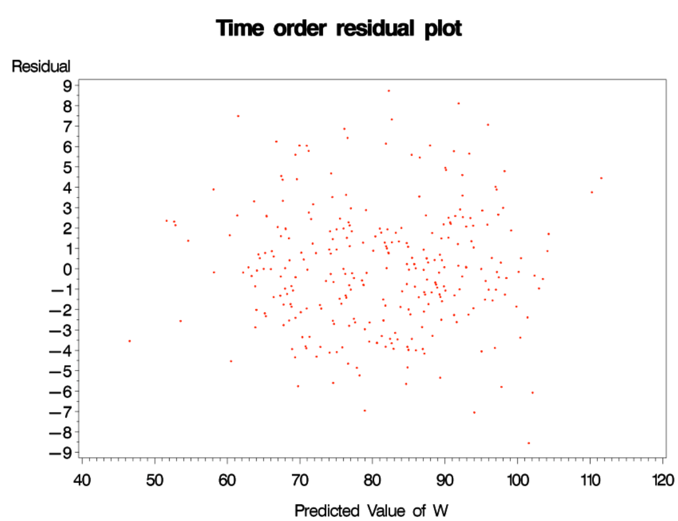

Figure 2.3 Time order plot for 1997-2006 full regression

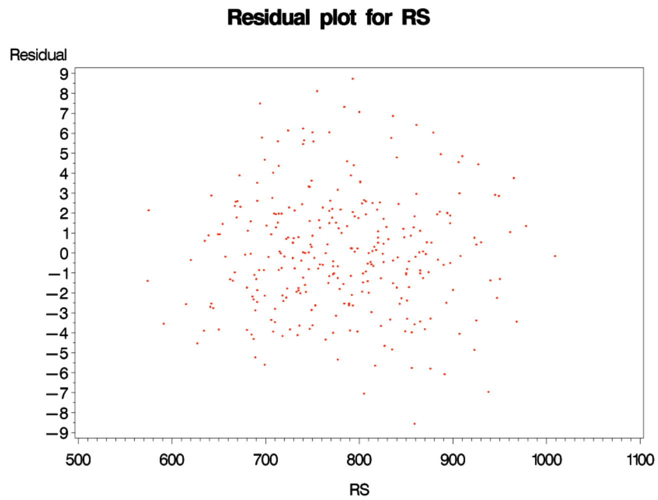

Figure 2.4 Runs scored residual plot for 1997-2006 full regression

As seen in Table 2.5, not all variables are significant. In fact, some, such as Fielding Percentage with a p-value of 0.9369, seem to have almost no predictive power. The model is also most likely overspecified to this particular data set and would not be as likely to do well on a new data set, thereby limiting the prediction power. A final problem with this initial method is the issue of multicollinearity in the regressors. Multicollinearity is when two or more regressors violate the assumption of linear independence and therefore, have correlation greater than zero between them. This violates one of our initial assumptions for linear regression. In order to address all of these issues, a method of reducing explanatory variables is needed. Numerous variable selection techniques were implemented and will be discussed in detail in the next section.

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	143.77906	491.54104	0.29	0.7701
RS	1	0.07459	0.00781	9.55	<.0001
H	1	0.04185	0.08724	0.48	0.6318
TB	1	-0.04558	0.05514	-0.83	0.4091
BB	1	0.01491	0.01253	1.19	0.2352
SO	1	-0.00481	0.00238	-2.02	0.0439
SB	1	0.00696	0.00589	1.18	0.2382
OBP	1	-30.07777	121.62517	-0.25	0.8049
SLG	1	290.46370	307.01927	0.95	0.3449
AVG	1	-196.46497	484.10240	-0.41	0.6852
ERA	1	-0.12823	0.16370	-0.78	0.4341
SV	1	0.37158	0.02933	12.67	<.0001
RA	1	-0.06654	0.00681	-9.77	<.0001
WHIP	1	-14.28841	12.49487	-1.14	0.2538
OAVG	1	-14.27854	12.49323	-1.14	0.2541
K_BB	1	0.55401	1.91199	0.29	0.7722
E	1	-0.02176	0.07713	-0.28	0.7781
DER	1	-57.20038	52.22308	-1.10	0.2743
FPCT	1	-37.77375	476.89818	-0.08	0.9369

Table 2.5 Full Regression Parameter Estimates

2.1.3 Variable Selection Methods

When a model is overspecified, we have the problem of selecting a subset of possible predictors while balancing conflicting objectives. We want to include all variables that have legitimate predictive skill and at the same time exclude all extraneous variables that reduce predictive skill and increase standard errors of regression coefficients. Ideally, we would be able to determine the single best subset of predictors to include, but this is not always easy to gauge. Several different algorithms can be used to produce different best subsets, and I explore four of the most popular methods in my project. In order to go more in depth, I implemented the variable selection techniques on each individual year from 1997-2006 to see which trends of subsets were visible [6].

2.1.3.1 Subset Selection Method

The first technique used was the subset variable selection method. Subset selection runs regressions with all subsets of the predictors and returns the R^2, adjusted R^2, and *Cp* statistics as measures of their accuracy in order to help choose the best possible subset. I focused on the results of the adjusted R^2 and the Cp statistic.

2.1.3.1.1 Adjusted R^2

Adjusted R^2 (R^2_{adj}) is a modification of R^2 that adjusts for the number of explanatory terms in a model. Unlike R^2, R^2_{adj} increases only if the new term improves the model more than would be expected by chance. R^2_{adj} will always be less than or equal to R^2. $R^2_{adj} = 1-(1-R^2)\frac{N-1}{N-P-1}$, where P is the total number of regressors in the model and N is the sample size [6].

The R^2_{adj} results are displayed in Table 2.6. The subset with the highest R^2_{adj} has been chosen for each year and the variables included are checked. Runs scored and saves are both included in every year's model. Runs allowed is chosen in 6 out of the 10 years. One of the problems with using R^2_{adj} is that it commonly picks out subsets that are larger than necessary, which might be why so many of the subsets still have around 10 variables.

	2006	2005	2004	2003	2002	2001	2000	1999	1998	1997
R	x	x	x	x	x	x	x	x	x	x
H	x		x	x					x	x
TB		x	x	x						x
BB		x		x			x		x	
SO	x	x		x						x
SB		x						x	x	x
OBP	x			x		x		x	x	
SLG		x	x	x						x
AVG			x						x	x
ERA		x	x						x	
SV	x	x	x	x	x	x	x	x	x	x
OR	x				x	x	x	x		x
WHIP	x			x	x			x		
OAVG		x		x					x	x
K_BB	x					x		x	x	
E	x		x					x		
DER	x			x						
FPCT								x	x	
	0.9524	**0.9497**	**0.9624**	**0.9258**	**0.9386**	**0.9662**	**0.9138**	**0.9654**	**0.9543**	**0.9053**

Table 2.6 R^2_{adj} Values and Subsets

2.1.3.1.2 Mallows' Cp statistic

The *Cp* statistic can be used as a subsetting criterion in selecting a reduced model without such problems. If *p* regressors are selected from a set of $k > p$, then $Cp = \frac{SSE_p}{s^2} - n + 2p$.

$SSE_p = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$ is the error sum of squares for the model with *P* regressors, $\hat{Y}_i$ is the i^th^ predicted value of Y_i from the *p* regressors, $s^2 = \sum_{i=1}^{n}\frac{(Y_i - \bar{Y})^2}{n}$ is the residual mean square after regression on the complete set of *k* regressors, and *n* is the sample size [6].

Cp is a measure of the "goodness of fit" for a model, and tends to be less dependent than R^2 on the number of effects in the model. Hence, *Cp* tends to find the best subset that includes only the important predictors of the dependent variable. Under a model not suffering from too much bias, $E(Cp) = E\left(\frac{SSE_p}{s^2}\right) - E(n) + 2E(p) \approx \frac{(n-p)\sigma^2}{\sigma^2} - n + 2p = p$. The *Cp* statistic generally picks out more reasonably sized subsets than R^2_{adj} and the smaller subsets seen here are much more of what is desired. Runs scored once again appears in all of the subsets, whereas saves and runs allowed are now only in 5 out of 10. ERA also appears in 5 out of 10 of the best subsets [6].

	2006	**2005**	**2004**	**2003**	**2002**	**2001**	**2000**	**1999**	**1998**	**1997**
R	x	x	x	x	x	x	x	x	X	x
H										
TB										
BB	x									
SO	x									
SB										
OBP		x								
SLG										
AVG										
ERA				x		x	X		x	x
SV	x	x	x			x		x	x	x
OR	x	x	x		x			x		
WHIP	x									
OAVG										
K_BB										
E		x								
DER		x						x		
FPCT										
	5.5493	**3.032**	**2.1322**	**3.9425**	**1.3524**	**-1.279**	**1.3524**	**3.2605**	**1.8188**	**0.6376**

Table 2.7 *Cp* Values and Subsets

2.1.3.2 Forward Selection Method

The simplest model building approach is called forward selection. In this approach, one adds variables to the model one at a time. At each step, each variable that is not already in the model is tested for inclusion in the model. The most significant of these variables is added to the model, so long as its P-value is below some pre-set level. This value is usually set above the conventional .05 level, because of the exploratory nature of this method [6].

Forward selection chooses runs scored and saves to be in every year's model and chooses runs allowed to be in 9 out of 10. The models chosen are still around 5 variables each, but this seems to be very strong evidence for using only runs scored, runs allowed, and saves in a model.

	2006	2005	2004	2003	2002	2001	2000	1999	1998	1997
R	x	x	x	x	x	x	x	x	x	x
H										x
TB										
BB			x				x			
SO										
SB	x									x
OBP	x	x			x	x				
SLG										
AVG			x							x
ERA		x	x							x
SV	x	x	x	x	x	x	x	x	x	x
OR	x	x	x	x	x	x	x	x	x	
WHIP	x	x		x	x	x				
OAVG					x					
K_BB								x		
E		x	x							
DER		x								
FPCT										

Table 2.8 Forward Selection Subsets

2.1.3.3 Backward Selection Method

Backward selection starts with fitting a model with all the variables. Then the least significant variable is dropped, so long as it is not significant at the chosen critical level. The model is successively re-fitted applying the same rule until all remaining variables are statistically significant [6].

Backward selection again shows strong evidence that runs scored and saves should be included in a more concise model for wins, but is less confident that runs allowed should be included. It could be that runs allowed becomes more significant when there are fewer variables in the model and that it was removed before this was allowed to transpire.

	2006	2005	2004	2003	2002	2001	2000	1999	1998	1997
R	x	x	x	x	x	x	x	x	x	x
H				x			x		x	x
TB		x		x						
BB		x							x	
SO	x	x								
SB									x	x
OBP	x							x	x	
SLG		x		x						
AVG	x		x				x		x	x
ERA		x	x				x		x	x
SV	x	x	x	x	x	x		x	x	x
OR	x				x	x		x		
WHIP				x						
OAVG		x		x						
K_BB	x					x		x		
E	x		x					x		
DER	x			x						
FPCT									x	

Table 2.9 Backward Selection Subsets

2.1.3.4 Stepwise Selection Method

Stepwise selection is a method that allows dropping or adding variables at the various steps. The process alternates between choosing the least significant variable to drop and then re-considering all dropped variables (except the most recently dropped) for re-introduction into the model. This means that two separate significance levels must be chosen for deletion from the model and for adding to the model. The second significance must be more strict than the first [6].

Stepwise selection poses a more compelling argument for the inclusion of runs allowed, runs scored, and saves as the only 3 explanatory variables necessary in the model for wins. In fact, a few of the years exclusively chose those 3 variables in this method.

	2006	2005	2004	2003	2002	2001	2000	1999	1998	1997
R	x	x	x	x	x	x	x	x	x	x
H										x
TB										
BB							x			
SO										
SB	x									x
OBP	x	x								
SLG										
AVG			x							x
ERA			x							x
SV	x	x	x	x	x	x	x	x	x	x
OR	x	x		x	x	x	x	x	x	
WHIP	x					x				
OAVG										
K_BB								x		
E			x							
DER										
FPCT										

Table 2.10 Stepwise Selection Subsets

2.1.4 Reduced Models and Results

From the results of the variable selection methods, I hypothesized that the most significant predictors in modeling wins are runs scored, runs allowed, and saves. These 3 variables are highlighted in Tables 2.6, 2.7, 2.8, 2.9, and 2.10 in order to show more clearly which selection models chose them.

A regression was run with these 3 explanatory variables on the 1997-2006 data to create a 10-year model. The result of the regression is very similar to that of the full model without the issues of overspecificity and multicollinearity. The greatest improvement is in the significance of the explanatory variables. As seen in Table 2.11, all variables in the model are now extremely significant.

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	59.56387	2.90364	20.51	<.0001
RS	1	0.09013	0.00235	38.37	<.0001
SV	1	0.39186	0.02999	13.07	<.0001
RA	1	-0.08315	0.00251	-33.14	<.0001

Table 2.11 1997-2006 RS/SV/RA Regression

In order to get more linear models for predicting the 2007 season, 2002-2006 data was regressed to create a 5-year model, and 2006 data was regressed to create a 1-year model. Some of the regression results are displayed in Table 2.12. These 3 models can now be used in making playoff picture predictions from regular season data.

	R^2	R^2_{adj}	MSE
10-year	0.9321	0.9314	9.99941
5-year	0.9272	0.9257	11.55597
1-year	0.9121	0.902	9.96633

Table 2.12 Regression Comparisons

2.2 Nonlinear Winning Percentage Models

Another way of approaching win prediction is through the use of sabermetric tools. Numerous different linear and nonlinear models have been created mostly through data analyses and implemented in different situations. I will study and apply three of the most famous nonlinear winning percentage predictors that are closely correlated with actual winning percentage. They all have the form $WP\% = \frac{RS^x}{RA^x + RS^x}$ and vary only in the choice of x.

2.2.1 Pythagorean Expectation

The Pythagorean Expectation is the most widely implemented nonlinear winning percentage model seen in common practice. Major League Baseball has even utilized it on its statistics website in order to give fans an expected wins and losses column along with the other core statistics. It was created by Bill James in 1981 to estimate how many games a baseball team should have won based on the number of runs they scored and allowed. The initial form was $WP\% = \frac{RS^2}{RS^2 + RA^2}$. The name of the formula comes from the fact that it "looks" like the Pythagorean Theorem, despite it having no relevant connection. Later data analyses by James and others led to a belief that 1.83 was a better exponent than 2 [1, 8].

Initially the correlation between the formula and actual winning percentage was simply an experimental observation, but a theoretical explanation can be derived.

Recall the 3-parameter Weibull distribution density:

$$f(x;\alpha,\beta,\gamma) = \begin{cases} \frac{\gamma}{\alpha}\left(\frac{x-\beta}{\alpha}\right)^{\gamma-1} e^{-\left(\frac{x-\beta}{\alpha}\right)^{\gamma}} & \text{if } x \geq \beta \\ 0 & \text{o.w.} \end{cases}$$

It is reasonable to assume that runs scored (*RS*) and runs allowed (*RA*) are from Weibull distributions because of the shape of the data [11]. Clearly, *RS* and *RA* cannot be completely independent because a game cannot end in a tie. But, Miller implemented chi-squared independence tests to show they still can fulfill this assumption [5].

To start, he broke the runs scored and allowed into bins $[0,1) \cup [1,2) \cup \ldots \cup [10,11) \cup [11,\infty)$. This gave an incomplete $r \times c$ two-dimensional contingency table with $(12-1)^2 - 12 = 109$ degrees of freedom. Thus, Miller had to use a modified χ^2 test for independence. First, let $Bin(k)$ denote the k$^{\text{th}}$ bin. For the 12 × 12 incomplete contingency table with these bins for both runs scored and allowed, the entry $O_{r,c}$ corresponded to the observed number of games where the team's runs scored is in $Bin(r)$ and the runs allowed are in $Bin(c)$. Because games can't end in a tie, $O_{r,r} = 0$ for all *r*. An iterative fitting procedure was then used to obtain maximum likelihood estimators for the $E_{r,c}$, the expected frequency of cell (r,c), under the assumption that if the runs scored and allowed are distinct, then the runs scored and allowed are independent.

The χ^2 statistic was then calculated by $\chi^2 = \sum_{r=1}^{n}\sum_{\substack{c=1 \\ c \neq r}}^{n} \frac{(O_{r,c} - E_{r,c})^2}{E_{r,c}}$ and was then compared to a critical value at the 95% threshold for the χ^2 with 109 degrees of freedom. The results showed that 13 out of 14 American League teams satisfied the assumption, therefore, it's logical to assume the independence of *RA* and *RS*. This lead to an important theorem involving the Pythagorean Expectation [5].

PE Theorem: Let *X* and *Y* be independent random variables from Weibull distributions with parameters $(\alpha_{RS}, \beta, \gamma)$ and $(\alpha_{RA}, \beta, \gamma)$, where α_{RS} and α_{RA} are chosen such that the means of *X* and *Y* are *RS* and *RA*. If $\gamma > 0$, then W-L%=$\frac{(RS-\beta)^{\gamma}}{(RA-\beta)^{\gamma} + (RS-\beta)^{\gamma}}$.

Proof: Let $X \sim Wei(\alpha_{RS}, \beta, \gamma)$ and $Y \sim Wei(\alpha_{RA}, \beta, \gamma)$. Then *RS*=$\alpha_{RS}\Gamma(1+\frac{1}{\gamma}) + \beta$ and *RA*=$\alpha_{RA}\Gamma(1+\frac{1}{\gamma}) + \beta$. Solving for the alphas, we obtain $\alpha_{RS} = \frac{RS-\beta}{\Gamma(1+\frac{1}{\gamma})}$ and $\alpha_{RA} = \frac{RA-\beta}{\Gamma(1+\frac{1}{\gamma})}$. (Refer to Appendix 5.2 for derivations of the Weibull distribution mean and variance.)
We need $P(X > Y)$.

$$P(X > Y) = \int_{\beta}^{\infty}\int_{\beta}^{x} f(x;\alpha_{RS},\beta,\gamma) f(y;\alpha_{RA},\beta,\gamma) dy dx$$

$$= \int_{\beta}^{\infty} \frac{\gamma}{\alpha_{RS}} \left(\frac{x-\beta}{\alpha_{RS}}\right)^{\gamma-1} e^{-\left(\frac{x-\beta}{\alpha_{RS}}\right)^{\gamma}} \int_{\beta}^{x} \left[\frac{\gamma}{\alpha_{RA}} \left(\frac{y-\beta}{\alpha_{RA}}\right)^{\gamma-1} e^{-\left(\frac{y-\beta}{\alpha_{RA}}\right)^{\gamma}} dy \right] dx$$

Let $\beta = 0$ as in the Pythagorean Expectation. Then,

$$P(X > Y) = \int_{0}^{\infty} \frac{\gamma}{\alpha_{RS}} \left(\frac{x}{\alpha_{RS}}\right)^{\gamma-1} e^{-\left(\frac{x}{\alpha_{RS}}\right)^{\gamma}} \int_{0}^{x} \left[\frac{\gamma}{\alpha_{RA}} \left(\frac{y}{\alpha_{RA}}\right)^{\gamma-1} e^{-\left(\frac{y}{\alpha_{RA}}\right)^{\gamma}} dy \right] dx$$

$$= \int_{0}^{\infty} \frac{\gamma}{\alpha_{RS}^{\gamma}} x^{\gamma-1} e^{-\left(\frac{x}{\alpha_{RS}}\right)^{\gamma}} \int_{0}^{x} \left[\frac{\gamma}{\alpha_{RA}^{\gamma}} y^{\gamma-1} e^{-\left(\frac{y}{\alpha_{RA}}\right)^{\gamma}} dy \right] dx$$

Using the substitution: $u = \left(\frac{y}{\alpha_{RA}}\right)^{\gamma}$; $du = \frac{\gamma}{\alpha_{RA}}\left(\frac{y}{\alpha_{RA}}\right)^{\gamma-1} = \frac{\gamma}{\alpha_{RA}^{\gamma}} y^{\gamma-1}$

$$P(X>Y)=\int_0^\infty \tfrac{\gamma}{\alpha_{RS}^\gamma} x^{\gamma-1} e^{-\left(\frac{x}{\alpha_{RS}}\right)^\gamma}\left[\int_0^{\left(\frac{x}{\alpha_{RA}}\right)^\gamma} e^{-u}du\right]dx$$

$$=\int_0^\infty \tfrac{\gamma}{\alpha_{RS}^\gamma} x^{\gamma-1} e^{-\left(\frac{x}{\alpha_{RS}}\right)^\gamma}\left[1-e^{-\left(\frac{x}{\alpha_{RA}}\right)^\gamma}\right]dx$$

Let $\dfrac{1}{\alpha^\gamma}=\dfrac{1}{\alpha_{RS}^\gamma}+\dfrac{1}{\alpha_{RA}^\gamma}$, then

$$P(X>Y)=\int_0^\infty \tfrac{\gamma}{\alpha_{RS}^\gamma} x^{\gamma-1} e^{-\left(\frac{x}{\alpha_{RS}}\right)^\gamma}\left[1-e^{-\left(\frac{x}{\alpha_{RA}}\right)^\gamma}\right]dx$$

$$=1-\tfrac{\alpha}{\alpha_{RS}^\gamma}\int_0^\infty \tfrac{\gamma}{\alpha}\left(\tfrac{x}{\alpha}\right)^{\gamma-1} e^{-\left(\frac{x}{\alpha}\right)^\gamma}dx$$

$$=1-\tfrac{\alpha}{\alpha_{RS}^\gamma}$$

$$=1-\tfrac{1}{\alpha_{RS}^\gamma}\tfrac{\alpha_{RS}^\gamma\alpha_{RA}^\gamma}{\alpha_{RS}^\gamma+\alpha_{RA}^\gamma}$$

$$=\tfrac{\alpha_{RS}^\gamma}{\alpha_{RS}^\gamma+\alpha_{RA}^\gamma}.$$

From here we resubstitute, and get $P(X>Y)=\frac{(RS-\beta)^\gamma}{(RA-\beta)^\gamma+(RS-\beta)^\gamma}$ □

The assumption of $\gamma>0$ is necessary for the formula to make sense in the real world. For example, if $\gamma=-.5$, $\beta=0$, *RS*=25, and *RA*=16, then $\frac{(RS-\beta)^\gamma}{(RA-\beta)^\gamma+(RS-\beta)^\gamma}=\frac{25^{-.5}}{25^{-.5}+16^{-.5}}=\frac{4}{9}<\frac{1}{2}$. Thus, when $\gamma<0$, a team that is scoring more runs than they allow is predicted to have a losing season which isn't reasonable.

From the derivation, we can determine the best value of $\gamma>0$ to create accurate results. Two methods were used to find this exponent. First, we used a least squares method. The general goal was to minimize the sum of squares error from the runs scored data plus the sum of squares error from runs allowed data. Let $\beta=-.5$, because discrete data is being modeled by a continuous function. Therefore, there are 3 free parameters: $\alpha_{RA},\alpha_{RS},\gamma$. We can now find the least squares estimate such that

$$\min_{\alpha_{RS},\alpha_{RA},\gamma}\left[\sum_{k=1}^{B} RS_{obs}(k)-G*A(\alpha_{RS},-.5,\gamma,k)^2+\sum_{k=1}^{B} RA_{obs}(k)-G*A(\alpha_{RA},-.5,\gamma,k)^2\right]^2$$

where k = bin number, B = number of bins, G = number of games and $A(\alpha,\beta,\gamma,k)$=area under the Weibull distribution for that number of runs.

A similar approach is taken in the method of maximum likelihood. We can find values of $\alpha_{RA},\alpha_{RS},\gamma$ that maximize the given likelihood function L, where

$$L(\alpha_{RS},\alpha_{RA},-.5,\gamma)=\binom{G}{RS_{obs}(1)\ldots RS_{obs}(B)}\prod_{k=1}^{B}A(\alpha_{RS},-.5,\gamma,k)^{RS_{obs}(k)}$$
$$\times\binom{G}{RA_{obs}(1)\ldots RA_{obs}(B)}\prod_{k=1}^{B}A(\alpha_{RA},-.5,\gamma,k)^{RA_{obs}(k)}.$$

Computationally, it is equivalent to maximize

$$\log\left[L(\alpha_{RS},\alpha_{RA},-.5,\gamma)\right]=\log\left[\binom{G}{RS_{obs}(1)\ldots RS_{obs}(B)}\right]+\log\left[\binom{G}{RA_{obs}(1)\ldots RA_{obs}(B)}\right]$$
$$+\sum_{k=1}^{B}RS_{obs}(k)\log\left[A(\alpha_{RS},-.5,\gamma,k)\right]+\sum_{k=1}^{B}RA_{obs}(k)\log\left[A(\alpha_{RA},-.5,\gamma,k)\right]$$

Thus, we can ignore the multinomial terms since they don't depend on the parameters, and find

$$\max_{\alpha_{RS},\alpha_{RA},\gamma}\left[\sum_{k=1}^{B}RS_{obs}(k)\log\left[A(\alpha_{RS},-.5,\gamma,k)\right]+\sum_{k=1}^{B}RA_{obs}(k)\log\left[A(\alpha_{RA},-.5,\gamma,k)\right]\right].$$

The results from Miller's analysis of the 1994 American league data helped confirm what Bill James' formula hypothesized. The Least Squares method resulted in a mean of $\gamma = 1.79$, with standard deviation 0.09. The Maximum Likelihood method gave a mean of $\gamma = 1.74$, with standard deviation 0.06. These means are reasonably close to James' exponent of 1.83 and thus, help confirm the validity of this method [5].

2.2.2 PythagenPort

The second winning percentage model that I will examine is the PythagenPort. The PythagenPort was created by Clay Davenport in 1999 as a further reaction to the Pythagorean Expectation. Instead of modeling runs scored and runs allowed with the Weibull distribution, he chose to model them with the discrete Poisson distribution. To build this model, we first need the likelihood of a team scoring *X* runs in a single game, given that they averaged *Y* runs per game [3].

Recall the Poisson distribution: $P(X,Y) = \frac{Y^X e^{-Y}}{X!}$, where *X* = number of events and *Y* = mean number of events. Certain factors led Davenport to believe that the Poisson was too narrow around the mean. Teams often score zero, and they score within a run of their average less often than predicted. Teams also play in different parks and under different circumstances each day. Thus, Davenport required a series of three Poisson equations to model teams run distribution. Each equation counted as one-third of their total and all are evaluated at the same value of *X*, but varying values of *Y*. Through data analysis, Davenport concluded the best *Y* values to match actual distributions were *RPG* and $RPG \pm 2\left(\frac{RPG}{4}\right)^{.75}$, where $RPG = \frac{RS+RA}{games}$. The function allows the difference around the mean to grow slowly as *RPG* increases. In order to test this, Davenport used a simulation. He used a random number generator to generate scores of 1620 games at a time, counted how many times team A outscored team B, and came up with the needed exponent, $x = \frac{\log(W/L)}{\log(RS/RA)}$, to satisfy the Pythagorean Expectation for the 1620-game sample. After generating these values, he ran a regression with the needed exponent as the response variable and *RPG* as the predictor. The exponent generated was $x = .45 + 1.5 * \log(RPG)$. Using *x* in $WP\% = \frac{RS^x}{RA^x + RS^x}$, we have a formula which is more accurate than the Pythagorean Expectation for the upper extreme environments, as it was tested in the 4-40 RPG range [3].

2.2.3 PythagenPat

The final nonlinear model examined in this project is the PythagenPat. It was developed by David Smyth and “US Patriot” as a response to the failure of the Pythagorean Expectation and PythagenPort when $RPG < 4$. The greatest improvement of this model over the others is its ability to produce an exponent of 1 at 1 *RPG*, which is a large factor in winning percentage predictions for low scoring teams. If a team played 162 games at 1 *RPG*, they would win each game they scored a run and lose each time they allowed a run. Therefore, to make $\frac{W}{W+L} = \frac{RS^x}{RA^x + RS^x}$ x must be set equal to 1. Using data sets and this initial condition, (1,1), along with the PythagenPort exponent at other points, the exponent, $x = (RPG)^{.287}$, was created. Again, this is used in the standard winning percentage formula, $WP\% = \frac{RS^x}{RA^x + RS^x}$. PythagenPort developer Clay Davenport called the PythagenPat, "simpler" and "more elegant" than his exponent and now prefers it [7].

Chapter 3: Prediction of the Playoffs

3.1 Playoff Picture Predictions

Comparing the linear models predictive power with that of the nonlinear models is not a simple task because there is no standard way. Thus, I created a simple squared error formula, $\sum_{i=1}^{n}\left(W_i - \hat{W}_i\right)^2$, where W_i is the actual wins, $\hat{W}_i$ is wins predicted by the model, and n is the number of teams. The comparison of the models is shown in Table 3.1. The deviations seem to be much larger in the nonlinear models, and are best for the 1 year model. This project is not necessarily concerned with how accurately wins are predicted, but more so with how accurately teams are predicted to make the playoffs. Thus, this error is not necessarily the most important thing, but can still be used to gauge somewhat the usefulness of the models.

$\sum_{i=1}^{30}(W_i - \hat{W}_i)^2$	**1 year**	**5 year**	**10 year**	**PE-2**	**PE-183**	**Pport**	**Ppat**
	358.96	375.20	366.00	558.76	503.86	529.02	1324.29

Table 3.1 2007 Error Comparison

The win predictions from the 3 linear models can be seen clearly in the Figure 3.2. They all seem to be fairly similar as far as how they predict. I have ordered the teams from most wins to least in order to make it more obvious where the model predicts well and where it fails. Even though a team like Boston is predicted to have about 6 more wins than they actually did, once again, it is more important that the model correctly identified them as a playoff team. The teams picked by the linear models to be in the playoffs are illustrated in Table 3.3. All three of the models made the same playoff predictions.

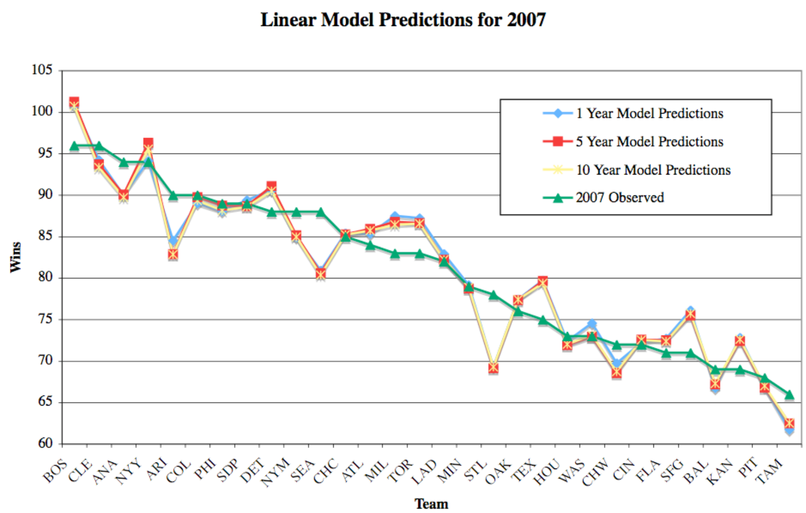

Figure 3.2 Linear Models Graph

Division	Actual Winner	Predicted 1 yr	Predicted 5 yr	Predicted 10 yr
ALE	BOS	BOS	BOS	BOS
ALC	CLE	CLE	CLE	CLE
ALW	ANA	ANA	ANA	ANA
AWC	NYY	NYY	NYY	NYY
NLE	PHI	PHI	PHI	PHI
NLC	CHC	*MIL*	*MIL*	*MIL*
NLW	ARI	*SDP*	*SDP*	*SDP*
NWC	COL	COL	COL	COL
Accuracy		75%	75%	75%

Table 3.3 Linear Model Predictions

The win predictions from the Pythagorean Expectation are seen in Figure 3.4. Both the accepted exponents of 1.83 and 2 are used to make the predictions. Their predictions seem to be almost identical. Overall, the graph looks very similar to the linear models. PythagenPort and

PythagenPat produce similar graphs as well, shown in Figures 3.5 and 3.6, respectively. PythagenPat doesn't look as accurate in that there seems to be a lot more deviance from the actual wins line.   Table 3.7 shows the nonlinear models predictions.

Overall, linear and nonlinear models did a relatively good job predicting the playoff picture. The National League West proved to be a hard division to predict because of the closeness of the teams in it. The Arizona Diamondbacks were a team that scored less runs than allowed which could have been the reason they were not picked to make the playoffs. Also, the San Diego Padres missed out on the playoffs by losing a one-game playoff with the Colorado Rockies, so it isn't unreasonable that they were picked. The National League East and National League Central were both hotly contested races that came down to the final weeks and even days, thus, making it almost impossible to predict accurately with a long term type of model. The success of the models could also be based upon using a full year's worth of runs scored, runs allowed, and saves.

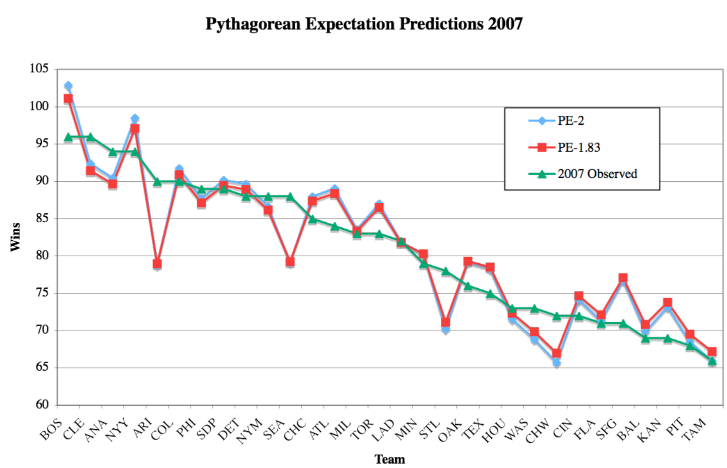

Figure 3.4 Pythagorean Expectation Graph

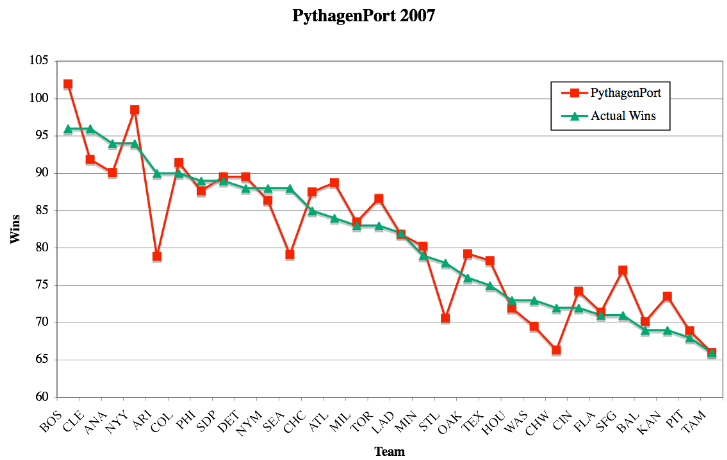

Figure 3.5 PythagenPort Graph

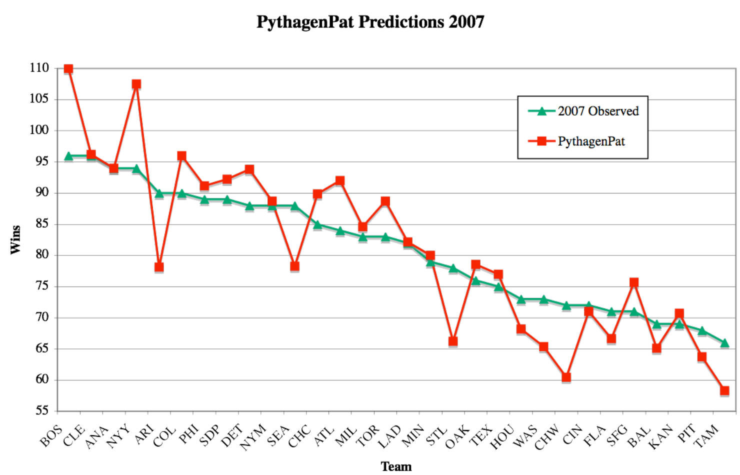

Figure 3.6 PythagenPat Graph

Division	Actual Winner	Predicted			
		PE2	PE183	Pport	Ppat
ALE	BOS	BOS	BOS	BOS	BOS
ALC	CLE	CLE	CLE	CLE	CLE
ALW	ANA	ANA	ANA	ANA	ANA
AWC	NYY	NYY	NYY	NYY	NYY
NLE	PHI	*ATL*	*ATL*	*ATL*	*ATL*
NLC	CHC	CHC	CHC	CHC	CHC
NLW	ARI	*SDP*	*SDP*	*SDP*	*SDP*
NWC	COL	COL	COL	COL	COL
Accuracy		*75%*	*75%*	*75%*	*75%*

Table 3.7 Pythagen Predictions

Predicting who makes the playoffs is not the same as predicting who wins when teams are in the playoffs. The long term regular season winning percentage no longer fully applies when teams play short series. Therefore, we need some sort of short term winning percentage to help predict the playoffs.

3.2 Log5 Formula

In order to determine a single game winning percentage, we need a new formula. The Log5 Formula, a 1981 Bill James invention, has had moderate success in predicting playoff games. It can give a reasonable prediction for the probability of team A winning a single game against team B [2].

The Log5 Formula is based upon 3 major assumptions. 1) Assume the Pythagorean Expectation is valid. 2) Assume all teams allow on average the same number of runs per game. 3) Assume that against each opponent a team will allow, on average, that opponent's average number of runs scored per game [2].

The Log5 Formula is as follows:

$WP_{AvB} = \frac{WP_A(1-WP_B)}{(WP_A)(1-WP_B)+(WP_B)(1-WP_A)}$ where

RS_A = Runs Scored per game by Team A, RS_B = Run Scored per game by Team B, RA_A = Team A Opponent Runs per game, RA_B = Team B Opponent Runs per game, RA_{League} = League Average Runs Allowed, WP_A = Regular Season Winning Percentage for Team A, and WP_B = Regular Season Winning Percentage for Team B.

In order to derive the Log5 Formula, we begin by determining the Pythagorean Expectation for each team in order to delegate their winning percentage [2].

$$WP_A = \frac{RS_A^{\,2}}{RS_A^{\,2}+RA_A^{\,2}} \qquad WP_B = \frac{RS_B^{\,2}}{RS_B^{\,2}+RA_B^{\,2}}$$

By assumption (2) we substitute RA_{League} for RA_A and RA_B:

$$WP_A = \frac{RS_A^{\,2}}{RS_A^{\,2}+RA_{League}^{\,2}} \qquad WP_B = \frac{RS_B^{\,2}}{RS_B^{\,2}+RA_{League}^{\,2}}$$

One can now solve both formulas for runs scored:

$$RS_A{}^2 = \frac{(WP_A)RA_{League}{}^2}{1-WP_A} \; (*) \qquad RS_B{}^2 = \frac{(WP_B)RA_{League}{}^2}{1-WP_B} \; (**)$$

By assumption (3), when team A plays team B, we can assume $RA_A = RS_B$ which leads to:

$$WP_{AvB} = \frac{RS_A{}^2}{RS_A{}^2 + RS_B{}^2}$$

Substituting $(*)$ and $(**)$ into the formula and simplifying, we obtain

$$WP_{AvB} = \frac{\frac{(WP_A)RA_{League}{}^2}{1-WP_A}}{\frac{(WP_A)RA_{League}{}^2}{1-WP_A} + \frac{(WP_B)RA_{League}{}^2}{1-WP_B}} = \frac{\frac{(WP_A)}{1-WP_A}}{\frac{(WP_A)}{1-WP_A} + \frac{(WP_B)}{1-WP_B}} = \frac{\frac{(WP_A)}{1-WP_A}}{\frac{(WP_A)(1-WP_B)+(WP_B)(1-WP_A)}{(1-WP_A)(1-WP_B)}} = \frac{(WP_A)}{1-WP_A}\frac{(1-WP_A)(1-WP_B)}{(WP_A)(1-WP_B)+(WP_B)(1-WP_A)}.$$

Thus, $WP_{AvB} = \frac{WP_A(1-WP_B)}{(WP_A)(1-WP_B)+(WP_B)(1-WP_A)}$ [2].

3.2 Using the Log5 method with the Negative Binomial Distribution

Recall the Negative Binomial distribution:

$f(k:r,p) = \binom{k+r-1}{k} p^r (1-p)^k$ where $p = P(\text{team A wins a game})$,
$r = \text{number of games to win the series}$, and $k = \text{team A losses in the series}$.

The Log5 method for determining single game winning probability can be used in conjunction with the negative binomial distribution to determine the probability of a team winning a series. Using the Log5 generated p for team A, the overall probability that team A wins can be determined. In the Divisional Series, a best-of-five series,

$P(\text{team A wins}) = p^3 + 3p^3(1-p) + 6p^3(1-p)^2$. In the League Championship Series and World Series, a best-of-seven series is used. In this setting,

$P(\text{team A wins}) = p^4 + 4p^4(1-p) + 10p^4(1-p)^2 + 20p^4(1-p)^3$. Table 3.8 has been generated with some selected p values to illustrate the probabilities for a team to winning a series.

p	Probability of winning a 5-game series	Probability of winning a 7-game series
0.4	0.3174	0.2898
0.44	0.3886	0.3706
0.48	0.4625	0.4563
0.52	0.5375	0.5437
0.56	0.6114	0.6294
0.6	0.6826	0.7102

Table 3.8 Probability of Winning a Series

3.3 Simulation with C

Using the method discussed above, I designed and wrote a simple simulation program in C seen in Appendix 5.4. It allows the user to specify the regular season winning percentage of two teams, the length of the series played between them, and the number of simulations to run. It then takes this information and computes the Log5 winning percentage for each team. Next it generates a random number between 0 and 1 to determine the winner of a game. This is repeated until one team has enough wins to win the series.

I used two different inputs for winning percentage and ran different numbers of simulations. The first input, $p = WP_{RS}$, where WP_{RS} is the regular season winning percentage for each team. The second input is $p = \left[1 - \frac{G_{AvB}}{162}\right]WP_{RS} + \frac{G_{AvB}}{162}WP_{AvB}$ where WP_{AvB} is regular season winning percentage of team A versus team B and G_{AvB} is games played by with team A versus team B. This second input takes the regular season meetings between the teams into account.

The results of the simulations can be seen in Tables 3.9 and 3.10. I ran 1000, 100, 10, and 1 simulations of each series in order to compare them. As the number of simulations gets higher, the number of series wins should go toward the Log5 winning percentage for each team. This isn't necessarily what transpires in the real world, thus, less simulations are probably more realistic. The results of the 2007 playoffs, regular season winning percentages, and regular season series between the playoff teams can be found in Appendix 5.5.

The simulation results give a slightly better insight into the playoffs. In Simulation I, the larger number of simulations predicted more accurately because the teams with higher winning percentages tended to win. In the case of the Cleveland vs. Boston series, the regular season win percentages were equal, so it was basically a toss up, and the simulation guessed wrong except in the 1- simulation case. Colorado was also predicted to lose in their League Championship Series by all but the 10-simulation case. Once again, the regular season winning percentages were extremely similar. Simulation II, which put more emphasis on the regular season games between the teams, predicted very accurately in the 1000-simulation case, except for Cleveland vs. New York. The regular season series between these teams was New York 6, Cleveland 0; thus, it had the largest impact on the winning percentage used in the simulation. Overall, the simulations show that no matter which team has a higher winning percentage, in a 1- simulation case, either team can come out a winner.

$p = WP_{RS}$								
DS	**BOS vs ANA**		**CLE vs NYY**		**ARI vs CHC**		**PHI vs COL**	
1000	*523*	477	*527*	473	*556*	444	493	*507*
100	46	*54*	*53*	47	*60*	40	47	*53*
10	5	5	4	*6*	5	5	5	5
1	*1*	0	0	*1*	0	*1*	0	*1*
LCS	**CLE vs BOS**		**ARI vs COL**			**WS**	**BOS vs COL**	
1000	*515*	485	*507*	493		**1000**	*584*	416
100	*51*	49	*53*	47		**100**	*57*	43
10	*6*	4	4	*6*		**10**	*7*	3
1	0	*1*	*1*	0		**1**	0	*1*

Table 3.9 Simulation I Results

$$p = \left[1 - \tfrac{G_{AvB}}{162}\right] WP_{RS} + \tfrac{G_{AvB}}{162} WP_{AvB}$$

DS	BOS vs ANA		CLE vs NYY		ARI vs CHC		PHI vs COL	
1000	*556*	444	435	*565*	*572*	428	476	*524*
100	*58*	42	43	*57*	*64*	36	*55*	45
10	5	5	4	*6*	5	5	5	5
1	*1*	0	0	*1*	0	*1*	0	*1*

LCS	CLE vs BOS		ARI vs COL	
1000	460	*540*	488	*512*
100	41	*59*	46	*54*
10	0	*10*	3	*7*
1	*1*	0	*1*	0

WS	BOS vs COL	
1000	*551*	449
100	*55*	45
10	4	*6*
1	0	*1*

Table 3.10 Simulation II Results

Chapter 4: Conclusions and Future Work

4.1 Results

The results that I got for the regular season were good overall. Each of the methods predicted 6 out of 8 playoff teams correctly with justifiable errors. The playoff simulation results were much less conclusive. Attempts with other multivariate techniques were made for predicting playoff results, but all failed to predict with even a 50% degree of accuracy. Therefore, this project is only a start on what could be used to predict the playoffs.

4.2 Secret Sauce

Further research would be necessary in order to more thoroughly predict the trends seen in playoffs. One such research area could be into the “secret sauce" formula created by Nate Silver. His research has found that a team pitching staff strikeout rate, the quality of a team’s defense, and the strength of a team’s closer are all very important components in champions.

In order to determine the validity of these assumptions, there needs to be quantitative measures for each. Strikeout rate is measured by Equivalent K/9 (EqK9), adjusted for a team's league and ballpark. Quality of defense is measured by the statistic Fielding Runs Above Average (FRAA), which is an estimate of the runs a defense has saved or cost its pitchers relative to the league average. Strength of closer is determined by Expectation Above Replacement (WXRL), which measures the wins the closer has saved versus what a replacement-level alternative would have done. To quote Silver, "In other words, teams that prevent the ball from going into play, catch it when it does and preserve late-inning leads are likely to excel in the playoffs." [9]

4.3 Final Thoughts

Predicting the course of a Major League Baseball season is never going to be an exact science. If we knew every outcome, there wouldn't be a point in playing the games. The goal of this project was to identify trends in winning teams over the past ten years in order to predict the future. Baseball is a sport in which numerous confounding variables come into play at the end of the season; thus, just predicting who will make the playoffs is a challenge. After that, as seen in the past, almost anything can happen, which is the true beauty of the game. In conclusion, in the words of the great player Ted Williams, "Baseball is the only field of endeavor where a man can succeed three times out of ten and be considered a good performer."

Chapter 5: Appendix

5.1 Sabermetric Definitions

Offensive

Batting Average = $\frac{\text{Hits}}{\text{At Bats}}$

Hits = total team hits

On Base Percentage = $\frac{\text{Hits+Walks+Hit By Pitch}}{\text{At Bats + Walks + Hit By Pitch + Sacrifice Flys}}$

Runs Scored = total team runs scored

Slugging Percentage = $\frac{\text{Total Bases}}{\text{At Bats}}$

Stolen Bases = total team stolen bases

Strikeouts = total team strikeouts

Total Bases = $\text{Singles} + 2\times\text{Doubles} + 3\times\text{Triples} + 4\times\text{Home Runs}$

Walks = total team walks

Pitching

ERA = Earned Run Average = $\frac{\text{Earned Runs}}{\text{Innings Pitched}}\times 9$

Opponent Averages = opponent batting average = $\frac{\text{Opponent Hits}}{\text{Opponent At Bats}}$

Runs allowed = total runs given up by pitching staff

Saves = total saves by pitching staff = When a pitcher 1) enters the game with a lead of three or fewer runs and pitches at least one inning, 2) enters the game with the potential tying run on base, at bat, or on deck, or 3) pitches three or more innings with a lead and is credited with a save by the official scorer

Strikeouts/Walk Ratio = $\frac{\text{Strikeouts Issued}}{\text{Walks Issued}}$

WHIP = $\frac{\text{Walks + Hits}}{\text{Innings Pitched}}$

Defensive

Defensive Efficiency Ratio = $\frac{\text{Batters Faced by Pitcher-Hits-Strikeouts-Walks-Hit By Pitch-Errors}}{\text{Batters Faced by Pitcher-Home Runs-Strikeouts-Walks-Hit By Pitch}}$

Errors = total team errors

Fielding Percentage = $\frac{\text{Assists + Put Outs}}{\text{Assists + Put Outs + Errors}}$

5.2 Weibull Distribution Mean and Variance Calculations

Recall the Γ-function, $\Gamma(s)=\int_0^\infty u^{s-1}e^{-u}du=\int_0^\infty u^{s}e^{-u}\frac{du}{u}$. Let $\mu_{\alpha,\beta,\gamma}$ denote the mean of $f(x;\alpha,\beta,\gamma)$ and $\sigma^2_{\alpha,\beta,\gamma}$ denote the variance. Using moment generating functions, we can determine $\mu_{\alpha,\beta,\gamma}$ and $\sigma^2_{\alpha,\beta,\gamma}$.

$E(X)=\mu_{\alpha,\beta,\gamma}=\int_0^\infty x\left(\frac{\gamma}{\alpha}\left(\frac{x-\beta}{\alpha}\right)^{\gamma-1}e^{-\left(\frac{x-\beta}{\alpha}\right)^{\gamma}}\right)dx$. Let $u=\left(\frac{x-\beta}{\alpha}\right)^{\gamma}$ and $du=\frac{\gamma}{\alpha}\left(\frac{x-\beta}{\alpha}\right)^{\gamma-1}dx$.

Then, solving for *x*,

$$u^{\frac{1}{\gamma}}=\frac{x-\beta}{\alpha}\Rightarrow \alpha u^{\frac{1}{\gamma}}=x-\beta\Rightarrow \alpha u^{\frac{1}{\gamma}}+\beta=x$$

Now, substituting $\alpha u^{\frac{1}{\gamma}}+\beta$ for x:

$$\begin{aligned}
\mu_{\alpha,\beta,\gamma} &=\int_0^\infty\left(\alpha u^{\frac{1}{\gamma}}+\beta\right)\left(e^{-u}\right)du\\
&=\alpha\int_0^\infty u^{\frac{1}{\gamma}}e^{-u}du+\beta\int_0^\infty e^{-u}du\\
&=\alpha\int_0^\infty u^{\frac{1}{\gamma}}e^{-u}du-\beta e^{-u}\Big]_0^\infty\\
&=\alpha\Gamma(1+\tfrac{1}{\gamma})+\beta
\end{aligned}$$

Similarly we find the second moment, $E(X^2)=\int_0^\infty x^2\left(\frac{\gamma}{\alpha}\left(\frac{x-\beta}{\alpha}\right)^{\gamma-1}e^{-\left(\frac{x-\beta}{\alpha}\right)^{\gamma}}\right)dx$. Once again, let $u=\left(\frac{x-\beta}{\alpha}\right)^{\gamma}$ and $du=\frac{\gamma}{\alpha}\left(\frac{x-\beta}{\alpha}\right)^{\gamma-1}dx$. Then,

$$\begin{aligned}
E(X^2) &=\int_0^\infty\left(\alpha u^{\frac{1}{\gamma}}+\beta\right)^2\left(e^{-u}\right)du\\
&=\int_0^\infty\left(\alpha^2u^{\frac{2}{\gamma}}+2\alpha\beta u^{\frac{1}{\gamma}}+\beta^2\right)\left(e^{-u}\right)du\\
&=\alpha^2\int_0^\infty u^{\frac{2}{\gamma}}e^{-u}du+2\alpha\beta\int_0^\infty u^{\frac{1}{\gamma}}e^{-u}du+\beta^2\int_0^\infty e^{-u}du\\
&=\alpha^2\Gamma(1+\tfrac{2}{\gamma})+2\alpha\beta\Gamma(1+\tfrac{1}{\gamma})+\beta^2
\end{aligned}$$

Now the variance can be easily determined.

$$\begin{aligned}
\sigma^2_{\alpha,\beta,\gamma} &=E(X^2)-\left[E(X)\right]^2\\
&=\alpha^2\Gamma(1+\tfrac{2}{\gamma})+2\alpha\beta\Gamma(1+\tfrac{1}{\gamma})+\beta^2\text{ - }\left(\alpha\Gamma(1+\tfrac{1}{\gamma})+\beta\right)^2\\
&=\alpha^2\Gamma(1+\tfrac{2}{\gamma})+2\alpha\beta\Gamma(1+\tfrac{1}{\gamma})+\beta^2-\alpha^2\left[\Gamma(1+\tfrac{1}{\gamma})\right]^2-2\alpha\beta\Gamma(1+\tfrac{1}{\gamma})-\beta^2\\
&=\alpha^2\Gamma(1+\tfrac{2}{\gamma})-\alpha^2\left[\Gamma(1+\tfrac{1}{\gamma})\right]^2
\end{aligned}$$

5.3 SAS Code

Full Regression Code

```
options linesize=80;
data baseball;
 infile 'allyears.csv' DLM=',' DSD MISSOVER FIRSTOBS=2;
 input Year$ Team$ League$ W RS H TB BB SO SB OBP SLG AVG ERA SV RA WHIP
OAVG K_BB E DER FPCT;
run;
proc reg;
 model W = RS H TB BB SO SB OBP SLG AVG ERA SV RA WHIP OAVG K_BB E DER
FPCT;
run;
```

Selection Methods Code

```
options linesize=80;
data baseball;
 infile 'allyears.csv' DLM=',' DSD MISSOVER FIRSTOBS=2;
 input Year$ Team$ League$ W R H TB BB SO SB OBP SLG AVG ERA SV OR WHIP
OAVG K_BB E DER FPCT;
run;
proc reg;
 model W=  R H TB BB SO SB OBP SLG AVG ERA SV OR OAVG WHIP K_BB E
DER FPCT/selection=rsquare ADJRSQ CP best=2;
proc reg;
 model W= R H TB BB SO SB OBP SLG AVG ERA SV OR OAVG WHIP K_BB E
DER FPCT/selection=f SLE=.2;
proc reg;
 model W= R H TB BB SO SB OBP SLG AVG ERA SV OR OAVG WHIP K_BB
E DER FPCT/selection=b SLS=.15;
proc reg;
 model W=R H TB BB SO SB OBP SLG AVG ERA SV OR OAVG WHIP K_BB E DER FPCT
/selection=stepwise SLE=.15;
run;
```

RS/RA/SV Regression Code

```
options linesize=80;
data baseball;
 infile 'allyears.csv' DLM=',' DSD MISSOVER FIRSTOBS=2;
 input Year$ Team$ League$ W RS H TB BB SO SB OBP SLG AVG ERA SV RA WHIP
OAVG K_BB E DER FPCT;
run;
proc reg;
 model W = RS SV RA;
 run;
```

5.4 C Simulation Code

```
/*   Project Sim      */
/*   Lindsey Dietz   */
/*   April 22, 2008  */

#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <time.h>

int main(){

char teamA[10];//team1 name
char teamB[10];//team2 name
float team1;//log5 team1 WP
float team2;//log5 team2 WP
int i=0;//loop variable
int l=0;//loop variable
int n=0;//number of simulations
double k=0.;
double team1wp,team2wp;
int team1_win,team2_win;
int serieslength;//length of series
int serieswin;//wins needed to win the series
int team1_count=0;
int team2_count=0;

printf("Enter team 1\n");
scanf("%s", teamA);
printf("Enter team 2\n");
scanf("%s", teamB);

printf("Enter %s regular season winning percentage\n",teamA);
scanf("%lf", &team1wp);
printf("Enter %s regular season winning percentage\n",teamB);
scanf("%lf", &team2wp);

printf("Enter series length\n");
scanf("%d",&serieslength);

printf("Enter the number of simulations\n");
scanf("%d",&n);

//Log5 initialization
team1=(team1wp*(1.-team2wp))/(team1wp*(1.-team2wp)+team2wp*(1.-team1wp));
team2=1-team1;
```

```
seriesswin=ceil((double)serieslength/2);

srand((unsigned)time( NULL ));
while(l<n){
        i=1;
        team1_win=0;
        team2_win=0;
        while (i<=serieslength && team1_win<serieswin){
                k=(double)rand()/((double)(RAND_MAX)+(double)(1));
                if (k<team1)
                        team1_win++;
                i++;
        }
        if (team1_win==serieswin)
                team1_count++;
        else
                team2_count++;
        l++;
}
        printf(" \033[2J");
        printf("\n\nIn %d simulations of a %d-game series\n",n,serieslength);
        printf("Team\t\t%s\t\t%s\n",teamA, teamB);
        printf("RS WP\t\t%lf\t%lf\n",team1wp,team2wp);
        printf("Log5 WP\t\t\%f\t%f\n",team1,team2);
      printf("Sim Series Wins\t%d\t\t%d\n", team1_count,team2_count);
      printf("Sim Series WP\t%.4f\t\t%.4f\n",
(float)team1_count/n,(float)team2_count/n);

return 0;

}
```

5.5 2007 Teams and Playoffs

Series	**Teams**	**Regular Season WP**	**Regular Season Wins**
Division Series	Anaheim Angels 0	0.58	4
	Boston Red Sox 3	0.593	6
Division Series	*Cleveland Indians* 3	0.593	0
	New York Yankees 1	0.58	6
Division Series	*Arizona Diamondbacks* 3	0.556	4
	Chicago Cubs 0	0.525	2
Division Series	*Colorado Rockies* 3	0.552	4
	Philadelphia Phillies 0	0.549	3
------------------	-----------------------------	----------------------	-----------------------------
League Series	Cleveland Indians 3	0.593	2
	Boston Red Sox 4	0.593	5
League Series	Arizona Diamondbacks 0	0.556	8
	Colorado Rockies 4	0.552	10
------------------	-----------------------------	----------------------	-----------------------------
World Series	Colorado Rockies 0	0.552	2
	Boston Red Sox 4	0.593	1

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